WORKED SOLUTION

STEP III 2016, Question 2

Pure • Normals to a Parabola

STEP III 2016, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q2

The key idea

Write down the normal at a general point $(at^2, 2at)$ and ask which values of t give a normal through P. That condition is a cubic in t, and one root, $t = p$, is always there (the normal at P passes through P). Removing it leaves a quadratic whose two roots are exactly q and r. From then on everything is driven by the sum and product of the roots, $q + r$ and qr.

Solution

Part (i)

On $y^2 = 4ax$ with $x = at^2$, $y = 2at$, we have $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2a}{2at} = \dfrac{1}{t}$, so the normal at parameter t has gradient $-t$:

$$y - 2at = -t\,(x - at^2), \qquad \text{that is} \qquad y + tx = 2at + at^3.$$

(This form also covers the vertex, $t = 0$, where the normal is the x-axis.)

The normal at t passes through $P(ap^2, 2ap)$ exactly when

$$2ap + tap^2 = 2at + at^3.$$

Dividing by $a > 0$ and rearranging,

$$\begin{aligned} 0 &= 2(p - t) + t(p^2 - t^2) \\ &= (p - t)(2 + t(p + t)) \\ &= (p - t)\,(t^2 + pt + 2). \end{aligned}$$

? **Ask yourself** Why can we throw away the factor $p - t$?

The root $t = p$ is the trivial one: the normal at P itself goes through P. We are told $Q \neq P$, so $q \neq p$, and the factor $p - q$ is non-zero. Hence

$$q^2 + qp + 2 = 0,$$

as required. By the same argument $r^2 + rp + 2 = 0$.

Part (ii)

So q and r are both roots of $t^2 + pt + 2 = 0$. Since $Q \neq R$ we have $q \neq r$, so they are its *two* roots, and

$$q + r = -p, \qquad qr = 2.$$

(In particular the discriminant is positive: $p^2 > 8$. Note also that $q = p$ would need $2p^2 + 2 = 0$, which is impossible, so Q and R are automatically different from P.)

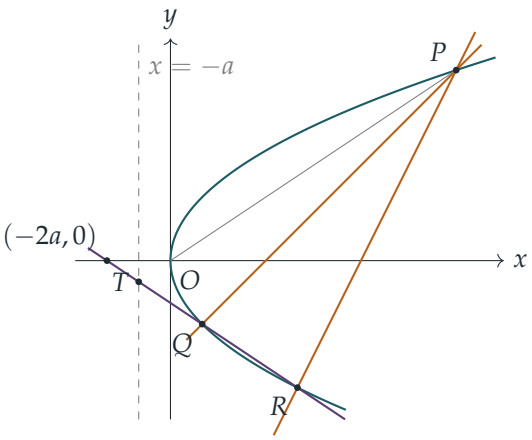
The chord through $(aq^2, 2aq)$ and $(ar^2, 2ar)$ has gradient $\dfrac{2a(q-r)}{a(q^2-r^2)} = \dfrac{2}{q+r}$, and its equation is

$$(q + r)\, y = 2x + 2aqr.$$

(Check: at Q the left side is $2aq^2 + 2aqr$, which equals the right side; similarly at R.) Substituting $q + r = -p$ and $qr = 2$:

$$-p\, y = 2x + 4a.$$

Whatever the value of p, the point $(-2a, 0)$ satisfies this equation. So QR always passes through $(-2a, 0)$, which does not depend on the choice of P.



The figure shows the case $a = 1$, $p = 3$, where $q = -1$ and $r = -2$: the two normals (amber) meet at P, the chord QR (plum) passes through $(-2, 0)$, and T lies on the dashed line $x = -1$.

Part (iii)

Since $p^2 > 8$, $p \neq 0$, so $P \neq O$ and the line OP is $y = \dfrac{2ap}{ap^2}\,x = \dfrac{2}{p}\,x$, or $x = \frac{1}{2}py$. (It is not parallel to QR, whose gradient is $-2/p$.)

Substituting $x = \frac{1}{2}py$ into the equation of QR:

$$-py = py + 4a \quad \Longrightarrow \quad y = -\frac{2a}{p}, \qquad x = \tfrac{1}{2}p\left(-\frac{2a}{p}\right) = -a.$$

So $T = (-a, -\frac{2a}{p})$, which always lies on the line $x = -a$ (the directrix of the parabola), independent of the choice of P.

The distance from the x-axis to T is $\dfrac{2a}{|p|}$. From part (ii), the existence of two distinct normals through P forces $p^2 > 8$, i.e. $|p| > 2\sqrt{2}$. Since $a > 0$,

$$\frac{2a}{|p|} < \frac{2a}{2\sqrt{2}} = \frac{a}{\sqrt{2}},$$

which completes the proof. The inequality is strict because $p^2 = 8$ would make $q = r$.

Checks

- In the figure's case $a = 1$, $p = 3$: the roots of $t^2 + 3t + 2$ are -1 and -2; the normals $y = x - 3$ and $y = 2x - 12$ both pass through $P(9, 6)$; the chord through $(1, -2)$ and $(4, -4)$ is $-3y = 2x + 4$, which passes through $(-2, 0)$; and $T = (-1, -\frac{2}{3})$, with $\frac{2}{3} < \frac{1}{\sqrt{2}} \approx 0.707$.
- Letting $|p| \to 2\sqrt{2}$ from above, Q and R merge and the height of T tends to $a/\sqrt{2}$, so the bound is sharp. A computer check over 20,000 random choices of a and p confirmed every result, with the height of T reaching 0.9996 of $a/\sqrt{2}$ but never exceeding it.

Common slips

- Forgetting to justify dividing by $p - q$: it is the condition $Q \neq P$ that allows it.
- Not using $q \neq r$ to say that q and r are *both* roots of the same quadratic, which is what gives $q + r = -p$ and $qr = 2$.
- Proving the final bound without saying where $p^2 > 8$ comes from: it is the requirement that the quadratic has two distinct real roots.

Beyond STEP

The cubic in t behind part (i) is a special case of a classical fact: for a general point (h, k) the condition is $at^3 + (2a - h)\,t - k = 0$, so from a point in the plane there are up

to three normals to a parabola, and the boundary between "one normal" and "three normals" is a curve called the *evolute* (here a semicubical parabola). Evolutes, envelopes and curvature are standard topics in a first course on the differential geometry of curves. Siklos, *Advanced Problems in Mathematics*, has more STEP problems in this coordinate-geometry style.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.